AI4Pain 2026 Challenge
NoPain / PainArm / PainHand Recognition

Two-Stage Physiological Pipeline

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Challenge Overview

AI4Pain 2026 — multimodal physiological pain recognition.

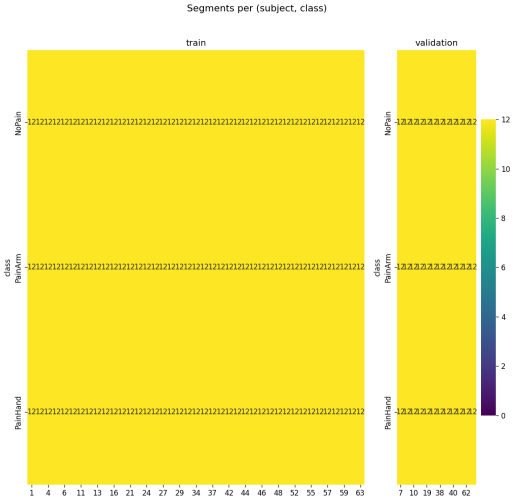
- **Task:** 3-class classification — `NoPain`, `PainArm`, `PainHand`.
- **Modalities:** BVP (blood volume pulse), EDA (electrodermal), Resp (respiration), SpO_2.
- **Sampling:** 4 channels per segment, 1000–1118 samples (median 1022).
- **Scoring:** macro-F1 / accuracy on held-out validation subjects.
- **Key constraint:** subject-disjoint splits $\Rightarrow$ LOSO evaluation required.

Objectives

1. Build a reliable 3-class pipeline that **beats single-modality baselines**.
2. Decouple **detection** (NoPain vs Pain) from **localization** (Arm vs Hand).
3. Explore **subject-level normalization** to beat cross-subject variance.
4. Explore **resampling** (truncate, linear, polyphase) for variable lengths.
5. Quantify **per-modality contribution** at each stage.
6. Deliver a reproducible `stage1 + stage2 + decoder` recipe.

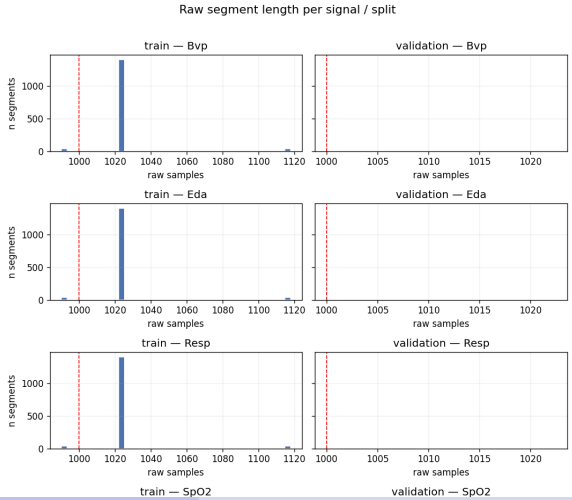
Dataset — Shape & Balance

- **Train:** $(1476, 4, 1000)$ — 41 subjects.
- **Validation:** $(432, 4, 1000)$ — 12 subjects.
- No subject overlap between splits.
- Every subject: exactly **12 / 12 / 12** (NoPain / PainArm / PainHand).
- Balanced at subject level $\Rightarrow$ enables exact-count decoding.

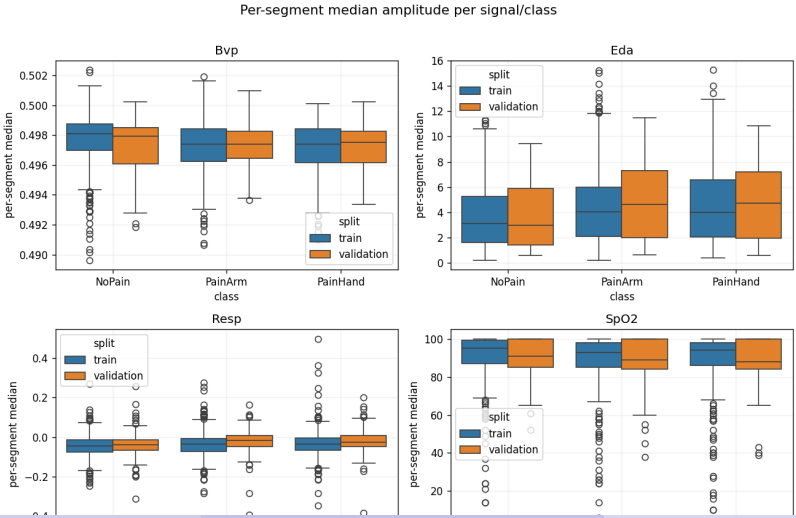


Dataset — Segment Lengths & Integrity

- Raw lengths: min 990, median **1022**, max 1118.
- Padding-NaN at tail: 36/1476 per signal (train).
- **No** interior NaN / Inf in any segment.
- **Constant SpO_2**: 254 train + 95 val segments (flat lines).
- SpO_2 has 349 byte-duplicate segments.
- Conclusion: SpO_2 is the weakest, most degraded channel.



Dataset — Signal Ranges



Baseline — Paper Modality Accuracies

Set	Modality	Accuracy
Validation	EDA	43.05 %
Validation	BVP	43.75 %
Validation	Resp	35.64 %
Validation	SpO_2	35.87 %
Validation	Multimodal	41.66 %
Test	EDA	–
Test	BVP	–
Test	Resp	–
Test	SpO_2	–
Test	Multimodal	–

Takeaway: paper baseline stuck near chance (33 %) for 3-class. Multimodal under-performed single best (BVP). Target: beat 43.75 %.

Our Approach — Two-Stage Pipeline

Architecture

Stage 1: binary NoPain vs Pain detector (BVP+EDA features, XGBoost).
Stage 2: binary PainArm vs PainHand localizer (Resp features, calibrated LogReg).
Decoder: joint-weighted exact-count (12/12/12) over (p1, p2).

- Stage 1 picks up **physiological arousal** (cardiovascular + sudomotor).
- Stage 2 exploits **respiration** as the best arm-vs-hand separator.
- Decoder enforces the known per-subject class distribution.

Our Approach — Default Best Config

Stage 1

- resample: `truncate1022`
- norm: `subject_robust`
- features: `bvp_eda_core` (98)
- model: `xgb`
- anchor: `center`, $\lambda = 0.5$

Stage 2

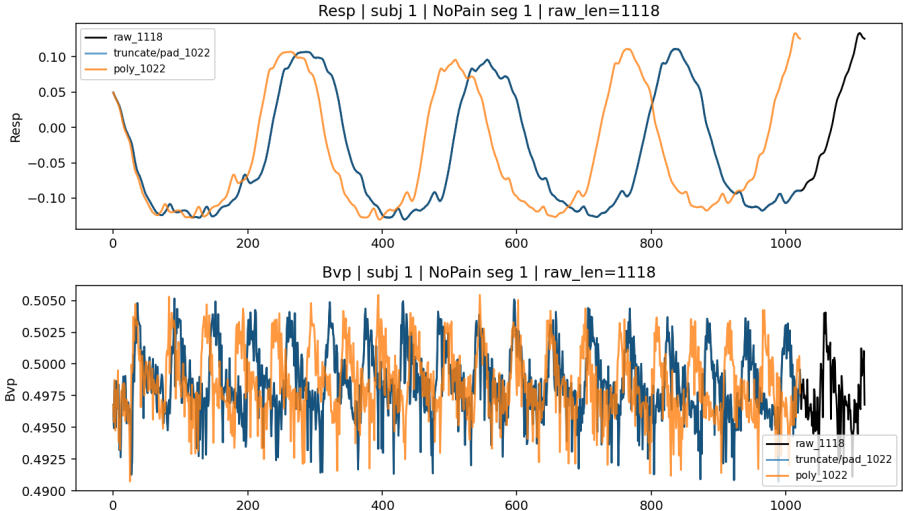
- resample: `truncate1022`
- norm: `subject_z`
- features: `resp_all` (44)
- model: `logreg`, `robust` scaler
- calibration: `isotonic`

Combined decoder: `joint_weighted` with w_0=0.8, w_1=1.2, w_2=1.4.

Preprocessing

- **Length alignment:** truncate / linear-resample / polyphase-resample to **1022** samples.
- **Padding handling:** drop tail-NaN before feature extraction.
- **Feature extraction:** 248 features per segment (time, frequency, morphology, Hjorth, entropy).
- **Subject-level normalization:**
  - `subject_z`: $(x - \mu_s)/\sigma_s$.
  - `subject_robust`: median / IQR per subject.
- **Feature subsets:** `bvp_eda_core` (98), `bvp_eda_resp_small`, `resp_all` (44), `all` (177–212).
- **Finding:** 1022 beats 1000; `truncate` beats linear/poly resampling overall.

Preprocessing — Resampling Example



Post-processing

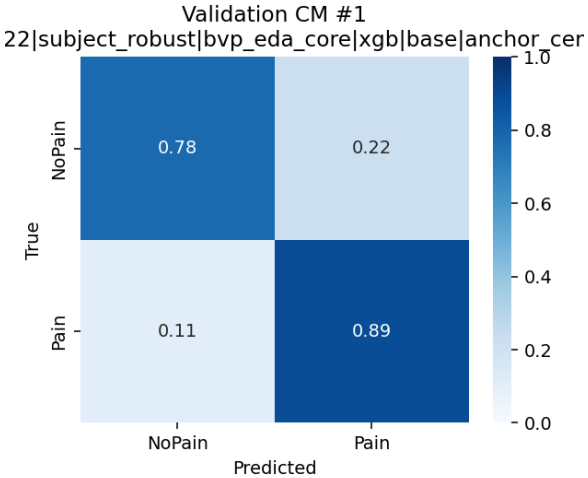
- **Probability calibration:**
  - Stage 1: `sigmoid` (Platt).
  - Stage 2: `isotonic` — essential for weighted decoding.
- **Anchor refinement (Stage 1):** pull scores toward the subject's `NoPain` centroid, $\lambda = 0.5$ (`anchor_center_l05`).
- **Decoders:**
  - `split_topk`: top-k per class, known counts.
  - `joint_weighted`: optimal assignment over joint (p1,p2) with class weights — *winner*.
- **Exact-count decoding:** enforces 12 / 12 / 12 per subject on validation.

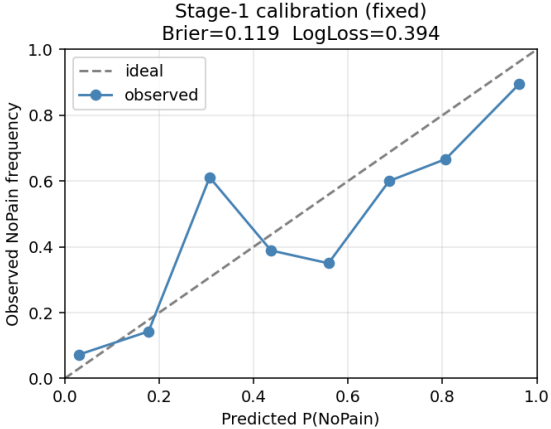
Stage 1 — Best Configs (Validation Macro-F1)

Config	Macro-F1	Acc	Prec NoPain	Rec NoPain
truncate1022 | subject_robust | bvp_eda_core | xgb | anchor_center_l05	**0.833**	**0.852**	0.778	0.778
truncate1022 | subject_robust | bvp_eda_core | xgb | base	0.828	0.847	0.771	0.771
poly1022 | subject_robust | bvp_eda_core | xgb | base	0.828	0.847	0.771	0.771
truncate1022 | subject_robust | bvp_eda_core | xgb | anchor_z_l05	0.828	0.847	0.771	0.771
truncate1022 | subject_z | bvp_eda_core | xgb | base	0.823	0.843	0.764	0.764
RBF SVM best (ref)	0.818	0.838	0.757	0.757

Winner: XGB on BVP+EDA with subject-robust norm + center anchor, $\lambda = 0.5$.

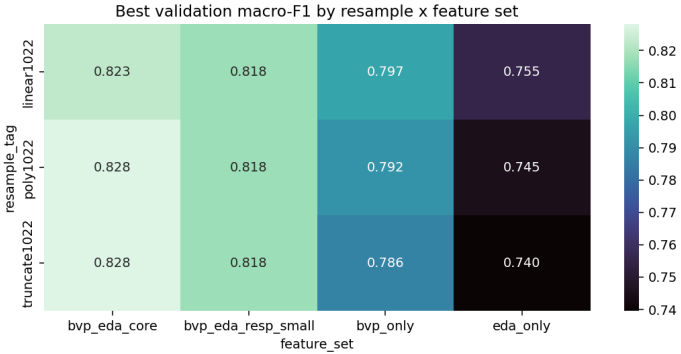
Stage 1 — Confusion & Calibration





Corrected calibration: predicted P(NoPain) vs observed NoPain freq. Brier = 0.119, LogLoss = 0.394, corr = 0.88.

Stage 1 — Resample × Feature-set Heatmap



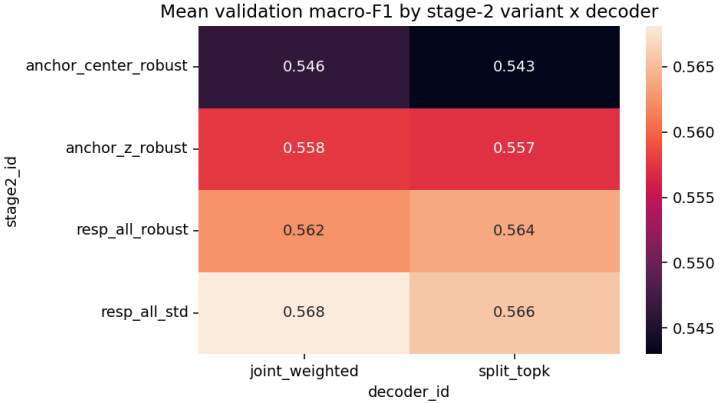
Stage 2 — Best Recipe

Winning stage-2 family

`subject_z` | `resp_all` | `logreg` | `robust` scaler | `isotonic` calibration.

- **Resp** carries the strongest arm-vs-hand localization signal.
- Larger model families (XGB, RF, SVM) did *not* reliably beat calibrated logreg.
- Isotonic calibration is **required** for weighted joint decoding to gain.
- Pairwise ranking and latent-intensity splits did not win.

Stage 2 — Decoder Heatmap



Stage 1 — Modality Table (Overall)

Modality	n_feat	val_acc	val_auc
BVP	52	0.806	0.858
EDA	46	0.755	0.829
RESP	44	0.606	0.580
SpO_2	35	0.657	0.642
EDA+BVP	98	**0.829**	**0.904**
EDA+BVP+RESP	142	0.810	0.900
All	177	0.819	0.905

Stage 1 — Modality Classwise

Modality	NoPain P	Pain P	NoPain R	Pain R	NoPain F1	Pain F1
BVP	0.708	0.854	0.708	0.854	0.708	0.854
EDA	0.632	0.816	0.632	0.816	0.632	0.816
RESP	0.410	0.705	0.410	0.705	0.410	0.705
SpO_2	0.486	0.743	0.486	0.743	0.486	0.743
EDA+BVP	**0.743**	**0.872**	0.743	0.872	0.743	0.872
EDA+BVP+RESP	0.715	0.858	0.715	0.858	0.715	0.858
All	0.729	0.865	0.729	0.865	0.729	0.865

Stage 2 — Modality Table (Overall)

Modality	n_feat	val_acc	val_auc
BVP	52	0.514	0.519
EDA	46	0.507	0.512
RESP	44	0.514	**0.566**
SpO_2	35	0.479	0.483
EDA+BVP	98	**0.549**	0.531
EDA+BVP+RESP	142	0.521	0.547
All	177	0.521	0.530

Resp holds the best ranking signal (AUC), EDA+BVP the best top-1 (acc).

Stage 2 — Modality Classwise

Modality	PainArm P	PainHand P	PainArm R	PainHand R	PainArm F1	PainHand F1
BVP	0.514	0.514	0.514	0.514	0.514	0.514
EDA	0.507	0.507	0.507	0.507	0.507	0.507
RESP	0.514	0.514	0.514	0.514	0.514	0.514
SpO_2	0.479	0.479	0.479	0.479	0.479	0.479
EDA+BVP	**0.549**	**0.549**	0.549	0.549	0.549	0.549
EDA+BVP+RESP	0.521	0.521	0.521	0.521	0.521	0.521
All	0.521	0.521	0.521	0.521	0.521	0.521

Combined 3-Class — Modality Overall

Modality	n_feat	val_acc	val_macro_f1
BVP	52	0.514	0.514
EDA	46	0.514	0.514
RESP	44	0.407	0.407
SpO_2	35	0.403	0.403
EDA+BVP	98	**0.558**	**0.558**
EDA+BVP+RESP	142	0.546	0.546
All	177	0.556	0.556

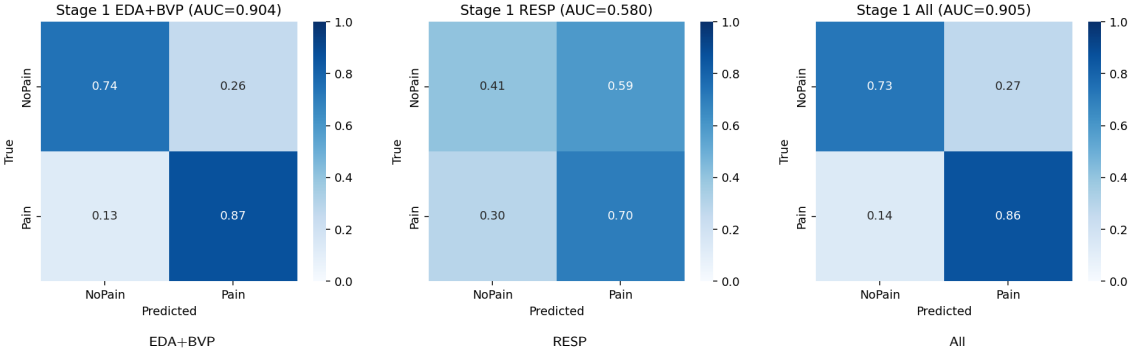
vs paper multimodal 41.66 % → **our 55.8 %** (+14 pts abs.).

Combined 3-Class — Classwise F1

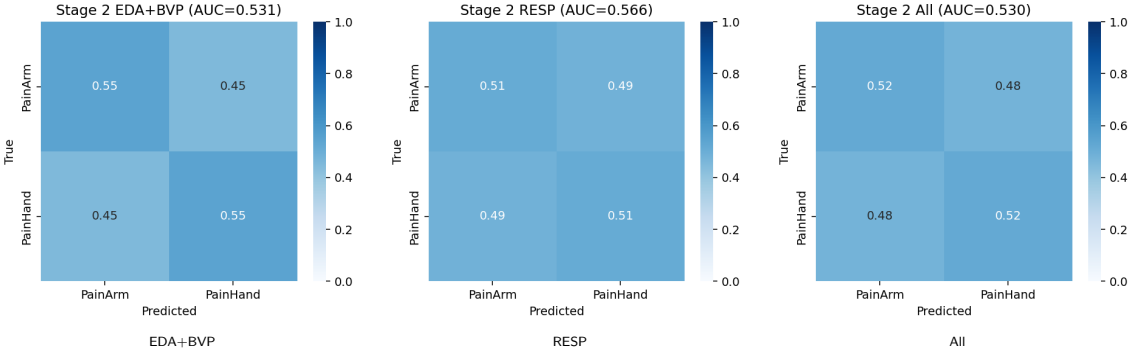
Modality	NoPain F1	PainArm F1	PainHand F1
BVP	0.701	0.438	0.403
EDA	0.646	0.403	0.493
RESP	0.424	0.340	0.458
SpO_2	0.507	0.368	0.333
EDA+BVP	**0.743**	**0.472**	0.458
EDA+BVP+RESP	0.715	0.444	0.479
All	0.743	0.451	0.472

NoPain easily separated. PainArm / PainHand remain the hard classes.

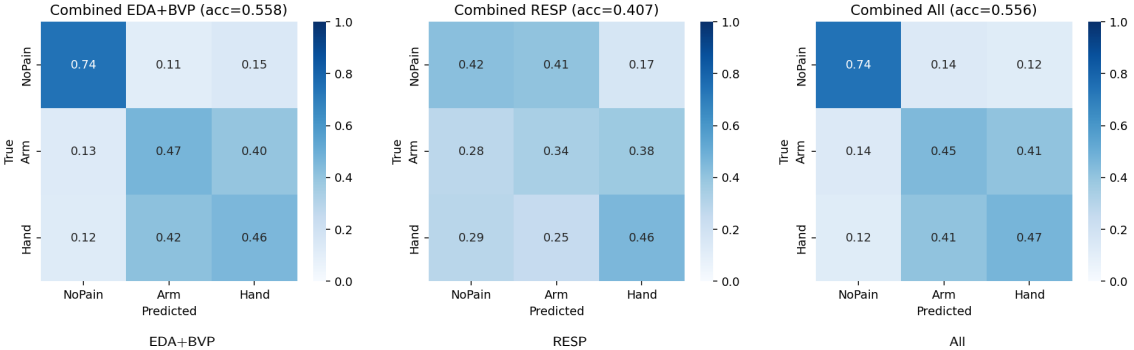
Modality Confusion Matrices — Stage 1



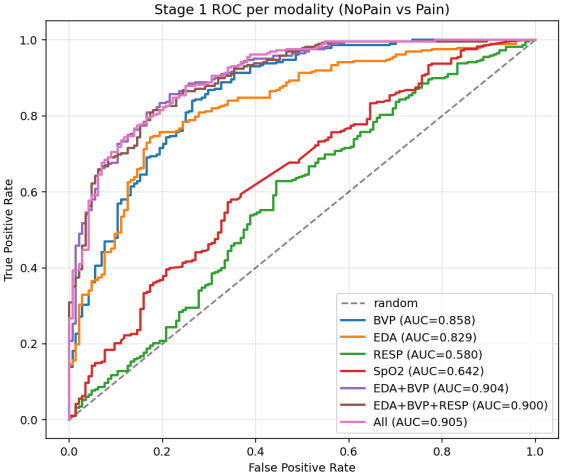
Modality Confusion Matrices — Stage 2

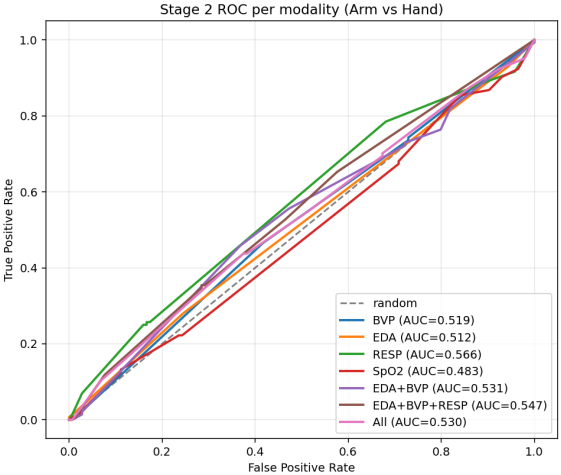


Modality Confusion Matrices — Combined



Stage 1 / Stage 2 ROC by Modality





Final Pipeline — Headline Numbers

Split	Accuracy	Balanced Acc	Macro-F1
train LOSO (41 subj)	0.558	0.558	0.558
validation (12 subj)	**0.604**	**0.604**	**0.604**

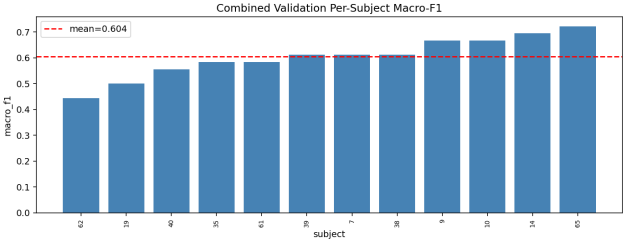
Best overall 3-class config:

`truncate1022` | `xgb_subject_z` | `resp_all_robust` | `joint_weighted`

macro-F1 = **0.6065** on validation.

- +18 pts over paper best modality (BVP 43.75 %).
- +19 pts over paper multimodal (41.66 %).

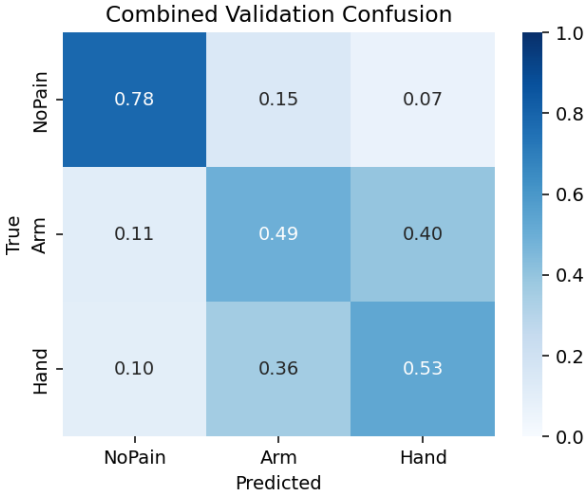
Final Pipeline — Per-Subject Spread



Per-subject accuracy on validation (12 subjects).

- Validation mean 0.588, std 0.084 (min 0.44, max 0.69).
- LOSO mean 0.579, std 0.111 (10/41 subjects below 0.5 — hard subjects).

Final Pipeline — Confusion Matrix



Grid-Optimized per Modality — Final Table

Modality	n_feat	acc	bal_acc	macro_f1
all	212	**0.549**	**0.549**	**0.549**
bvp	62	0.512	0.512	0.512
eda	55	0.512	0.512	0.512
bvp+eda	117	0.507	0.507	0.507
resp	53	0.410	0.410	0.410
spo2	42	0.396	0.396	0.396

After *per-modality* grid search, `all`-features wins on validation; `bvp_eda` drops slightly under per-modality tuning (regression: overfit selection).

What Helped

1. **1022**-length windows (vs 1000).
2. Exact **12/12/12** subject-level decoding.
3. **Calibrated + weighted** joint decoding.
4. Upgraded stage-1 detector with **BVP+EDA** core features.
5. Subject-level **robust** normalization.
6. **NoPain-anchor** refinement, $\lambda = 0.5$, center.

What Did Not Win

1. Latent intensity split.
2. Pairwise-ranking stage 2.
3. Heavy stage-2 reformulations (MoE, responder gating).
4. Linear / polyphase resampling vs simple truncation.
5. Replacing XGBoost stage-1 with SVM (RBF best = 0.818, still under 0.833).

Conclusion

- Two-stage **detect-then-localize** beats one-shot 3-class.
- **BVP+EDA** for detection, **Resp** for localization — modality split is physiologically sensible.
- Calibration + exact-count decoding closes most of the gap to validation ceiling.
- **Final:** macro-F1 = 0.606 (val), vs paper multimodal 0.417.
- Remaining ceiling: PainArm vs PainHand on hard subjects — per-subject diagnostics suggest adaptation, not bigger models.

Thank you.

Questions?